MIL-C-25974D(USAF)
1 April 1983
Superseding
MIL-C-25974C(USAF)
10 February 1978

MILITARY SPECIFICATION

CONVERTER, LIQUID OXYGEN GCU-10/A

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of liquid oxygen converter with a capacitance gaging element. The converter provides storage and pressurization of 10 liters of liquid oxygen to 300 pounds per square inch gage (psig) nominal pressure. The converter does not include warming coils. The converter is designated Type GCU-10/A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.2 Specifications and standards. The following specifications and standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	-	Box, Shipping, Fiberboard
PPP-T-60	-	Tape, Packaging, Waterproof

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-V-9050	-	Valve, Oxygen, Aircraft, Pressure Relief
MIL-C-25516	-	Connector, Electrical, Miniature, Environmental Resistant Type, General Specification for
MIL-C-25666	-	Converter, Liquid Oxygen, Capacitance Type Gaging, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-794	-	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831	-	Test Reports, Preparation of
MS33656	-	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. Converters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 General specification. The requirements of MIL-C-25666 shall apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.3 Design. The converter shall conform to the dimensions specified in figures 1, 2, and 3. Port and electrical connector locations and angles shall be as specified in table I. Reference planes in table I are defined in figure 4. The converter shall have a minimum capacity of 10 liters of liquid oxygen measured at 1 atmosphere, plus at least 3 percent of this volume to allow for the change in liquid density at the operating pressure. The converter shall be designed for an operating pressure of 300 psig.

3.3.1 Schematic diagram. The converter shall be designed to withstand the following "g" forces without damage:

Vertical (down)	14g
Vertical (up)	6g
Lateral	6g

3.3.3 Relief valves. The relief valves shall conform to the performance requirements of types III and IV of MIL-V-9050 and shall be of such size and shape to fit the over-all dimensions specified for the converter.

3.3.4 Weight. The weight of the complete converter shall not exceed 18 pounds.

3.3.5 Marking of ports. Ports shall be marked in accordance with figure 1.

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3.4 Performance.

3.4.1 Permanent volumetric expansion. The pneumatic or hydrostatic pressure used for the permanent volumetric expansion test shall be 500 psig.

3.4.2 Evaporation loss. The evaporation loss of the converter shall not exceed 3 pounds of oxygen when the converter is tested as specified.

3.4.3 Final assembly pressure. The pressure used for the final assembly pressure test shall be 320 psig.

3.4.4 Filling time. Filling time shall not exceed 10 minutes when the converter is tested as specified.

3.4.5 Buildup time. Buildup time shall not exceed 10 minutes when the converter is tested as specified.

3.4.6 Calibration. The capacitance sensing element shall have characteristics which provide for a linear relationship between electrical capacitance and volume of liquid oxygen sensed. The relationship shall be in accordance with figure 6, using the equations that are specified in pagagraph 3.4.6.1 to determine tolerances. An example of the use of these equations is as follows: If the actual quantity of liquid oxygen at the 6 liter test point is 6.1 liters, the measured capacitance must be between C maximum and C minimum.

C minimum = 5.7 x 6.1 + 122.3 = 157.3 pF
C maximum = 5.9 x 6.1 + 124.7 = 160.7 pF

3.4.6.1 Equations. The following equations shall be used to calculate the maximum and minimum capacitances.

Values less than 2 liters	Values greater than 2 liters
C minimum = 5.6L + 122.5	C minimum = 5.7L + 122.3
C maximum = 6.0L + 124.5	C maximum = 5.9L + 124.7

C = Capacitance in picofarad (pF)
L = Exact quantity of liquid oxygen in liters (L)

3.4.7 Delivery rate. The converter shall deliver constant gas flows up to 80 liters of gaseous oxygen per minute for not less than 1 hour with the operating pressure remaining between 295 and 315 psig after the first 15 minutes of operation. An intermittent increase or decrease of pressure by an additional 5 psig is allowable provided these variances do not exceed a 2-minute interval for any one pressure cycle.

3.4.8 Extreme temperature operation . The operating pressure shall remain between 295 and 320 psig after the first 15 minutes of flow at the low temperature and between 290 and 315 psig after the first 15 minutes of flow at the high temperature. An intermittent increase or decrease of pressure by an additional 5 psig is allowable provided these variances do not exceed a 2-minute interval for any one pressure cycle.

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3.4.9 Vibration. After the converter has been vibrated in accordance with L-C-25666, the evaporation loss of the converter in the buildup condition all not exceed 5 pounds of oxygen.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the ntract or purchase order, the supplier is responsible for the performance of l inspection requirements as specified herein. Except as otherwise ecified in the contract or order, the supplier may use his own or any other cilities suitable for the performance of the inspection requirements ecified herein, unless disapproved by the Government. The Government serves the right to perform any of the inspections set forth in the ecification where such inspections are deemed necessary to assure supplies d services conform to prescribed requirements.

4.2 Classification of inspection. The inspection required by this ecification is classified as follows:

(a) Quality conformance inspection
(b) Qualification inspection

4.3 Quality conformance inspection. The quality conformance inspection all be in accordance with MIL-C-25666.

4.4 Qualification inspection. The qualification inspection shall consist all of the methods of inspections that are described in MIL-C-25666.

4.4.1 Samples. At least two converters shall be subjected to the alification inspection. One converter shall be submitted to the qualifying tivity (see 6.3). The converter that is submitted to the qualifying tivity may be a tested converter provided it has not been subjected to the nfire test that is described in MIL-C-25666.

4.4.2 Report. Three copies of a qualification inspection report that has en prepared in accordance with MIL-STD-831 shall be submitted to the alifying activity (see 6.3).

4.5 Packaging inspection.

4.5.1 Quality conformance. The inspection of the preservation-packaging d interior marking shall be in accordance with Group A and B quality nformance inspection requirements, Section 4 of MIL-P-116. The sampling and spection of the packing and marking for shipment and storage shall be in cordance with the quality assurance provisions of the applicable container ecification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be in accordance th MIL-STD-794, level A or C, as specified by the procuring activity.

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5.1.1 Level A. Each converter pressurized to a maximum of 25 PSI shall be preserved in accordance with method III of MIL-P-116. The unit container shall conform to PPP-B-636, type C/F, style RSC, Class V3c. The closure shall be accomplished in accordance with the procedure specified in appendix of the applicable specification.

5.1.2 Level C. Each unit shall be packaged in a manner that will prevent physical damage during transit from the supply source to the first receiving activity for immediate use.

5.2 Packing. Unless otherwise specified by the procuring activity, the items shall be provided packing which will afford adequate protection against damage during shipment, handling, intermediate storage and world-wide redistribution at lowest cost.

5.2.1 Level A. Items packaged as specified in 5.1.1 shall be packed in fiberboard containers conforming to PPP-B-636, class weather reisitant, style RSC, grade V3s or V3c. As far as practical, exterior containers shall be uniform in shape and size, be of minimum cube and tare consistant with the protection required, and contain identical quantities. Closure and water--proofing shall be accomplished by sealing all seams corners and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60. Reinforcement requirements shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding.

5.2.2 Level B. Items packaged as specified in 5.1.1 shall be packed in fiberboard containers conforming to style RSC, class domestic, minimum grade 200 of PPP-B-636. Containers shall be of uniform shape and size and contain identical quantities. The container closure shall be accomplished in accordance with Method II, 30.3.1.2 of the appendix to PPP-B-636.

5.2.3 Level C. The packaged items shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity at lowest cost. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking. All unit and shipping containers shall be marked in accordance with MIL-STD-129. In addition, the following precautionary marking shall appear on the unit and shipping container:

CAUTION OXYGEN COMPONENTS - ALL OIL, GREASE, SHOP RESIDUE, OR OTHER CONTAMINANTS HAVE BEEN REMOVED. DO NOT OPEN UNTIL READY FOR USE.

6. NOTES

6.1 Intended use. The converter covered by this specification is intended for use on F-104B aircraft.

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6.2 Ordering data . Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Selection of applicable levels of preservation and packaging (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they proposed to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for this Qualified Products List is Oklahoma City Air Logistics Center/MMEDO, Tinker Air Force Base OK 73145 and information pertaining to qualification of products may be obtained from that activity.

6.3.1 Qualification test data. When requested, the manufacturer shall submit engineering drawings and test reports in accordance with Defense Standardization Program Document SD-6, Provisions Governing Qualification.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Preparing Activity:
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Project Number:
1660-F431

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TABLE 1 - PORT AND ELECTRICAL CONNECTOR LOCATIONS AND ANGLES

FILL PORT

A. Size and type of threads
1/4 ANPT internal

B. Location of center point of opening
(1) 1-1/8 inches right of Reference Plane A
(2) 5-7/16 inches in front of Reference Plane B
(3) 4-3/4 inches below Reference Plane C

C. Direction of boss certerline specified as projection on reference plane
(1) 40 degrees downward and forward in Reference Plane A
(2) 13 degrees downward toward right in Reference Plane B
(3) 13 degrees forward toward right in Reference Plane C

SUPPLY PORT

A. Size and type of threads
MS33656-5

B. Location of center point of opening
(1) 1.915 inches left of Reference Plane A
(2) 5-3/4 inches in front of Reference Plane B
(3) 3-1/2 inches below Reference Plane C

C. Direction of boss certerline specified as projection on reference plane
(1) A point in Reference Plane A
(2) Horizontal line in Reference Plane B
(3) Horizontal line in Reference Plane C

BUILD-UP PORT

A. Size and type of threads
1/8 ANPT internal

B. Location of center point of opening
(1) 1/2 inch right of Reference Plane A
(2) 6-9/16 inches in front of Reference Plane B
(3) 4-11/32 inches above Reference Plane C

C. Direction of boss centerline specified as projection on reference plane
(1) 25 degrees upward and forward in Reference Plane A
(2) 45 degrees upward toward right in Reference Plane B
(3) 60 degrees forward toward right in Reference Plane C

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TABLE I - PORT AND ELECTRICAL CONNECTOR LOCATIONS AND ANGLES (CONT)

VENT PORT

A. Size and type of threads
   3/8 ANPT internal
B. Location of center point of opening
   (1) 2-11/16 inches left of Reference Plane A
   (2) 6-1/32 inches in front of Reference Plane B
   (3) 1-1/4 inches above Reference Plane C

C. Direction of boss centerline specified as projection on reference plane
   (1) 10 degrees upward and forward in Reference Plane A
   (2) 10 degrees upward toward left in Reference Plane B
   (3) 25 degrees forward toward left in Reference Plane C

LOW PRESSURE RELIEF PORT

A. Size and type of threads
   1/8 ANPT internal

B. Location of center point of opening
   (1) 1-1/16 inches right of Reference Plane A
   (2) 6 inches in front of Reference Plane B
   (3) 3-3/16 inches above Reference Plane C

C. Direction of boss centerline specified as projection on reference plane
   (1) 30 degrees downward and forward in Reference Plane A
   (2) Vertical line in Reference Plane B
   (3) 90 degrees forward and back in Reference Plane C

HIGH PRESSURE RELIEF PORT

A. Size and type of threads
   1/8 ANPT internal

B. Location of center point of opening
   (1) 2-5/32 inches left of Reference Plane A
   (2) 5-3/4 inches in front of Reference Plane B
   (3) 3 inches above Reference Plane C

C. Direction of boss centerline specified as projection on reference plane
   (1) 35 degrees downward and forward in Reference Plane A
   (2) Vertical line in Reference Plane B
   (3) 90 degrees forward and back in Reference Plane C

HIGH IMPEDENCE GAGE CONNECTOR

A. Size and type of threads
   MIL-C-25516, B Polarity

B. Location of center point of opening
   (1) 2-5/8 inch right of Reference Plane A
   (2) 4-31/32 inches in front of Reference Plane B
   (3) 5-1/16 inches above Reference Plane C

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C. Direction of boss centerline specified as projection on reference plane
   (1) Horizontal line in Reference Plane A
   (2) Horizontal line in Reference Plane B
   (3) 30 degrees forward toward right in Reference Plane C

LOW IMPEDENCE GAGE CONNECTOR

A. Size and type of threads
   MIL-C-25516, A Polarity

B. Location of center point of opening
   (1) 2-5/8 inches right of Reference Plane A
   (2) 4-31/32 inches in front of Reference Plane B
   (3) 4-11/32 inches above Reference Plane C

C. Direction of boss centerline specified as projection on reference plane
   (1) Horizontal line in Reference Plane A
   (2) Horizontal line in Reference Plane B
   (3) 30 degrees forward toward right in Reference Plane C

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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